

Mass



1 Convert the following:

a 60 g to milligrams (mg)

c 87 mg to grams (g)

e 946 g to kilograms (kg)

g $3\frac{3}{5}$ kg to grams (g)

i 630 kg to tonnes (t)

k 0.493 t to kilograms (kg)

b 0.47 g to milligrams (mg)

d 8505 mg to grams (g)

f 6.251 kg to grams (g)

h 8500 kg to tonnes (t)

j 9.272 t to kilograms (kg)

2 Valerie has fruit everyday for breakfast. Today she consumed 45 cherries, which weighed a total of 180 grams. On average, how many milligrams does one cherry weigh?

3 True or False: 5 kg > 5600 g

4 What metric unit represents 'one thousandth of a kilogram'?

5 Aoife's bag weighs $\frac{1}{2}$ of a kilogram. How many grams does the bag weigh?

6 The cost of mining gold for a mining company is \$840 per gram. How much will it cost the company to mine 0.2 kilograms of gold?

Length

7 State whether the following units can be used to measure length:

a Litres, L

c Cubic centimetres, cm³

e Square meters, m²

b Centimetres, cm

d Kilometres, km

f Millimetres, mm

8 Convert the following lengths, writing your answer as a decimal:



- | | |
|---|---|
| a 80 mm to centimetres (cm) | b 59 mm to centimetres (cm) |
| c 4 m to centimetres (cm) | d 9.36 m to centimetres (cm) |
| e $6\frac{3}{4}$ m to centimetres (cm) | f 9000 m to kilometres (km) |
| g 4832.1 m to kilometres (km) | h 6.52 m to millimetres (mm) |
| i 49.1 cm to metres (m) | j 1402 cm to metres (m) |
| k 7.66 km to metres (m) | l $\frac{4}{5}$ km to metres (m) |

9 A skyscraper is 961 metres tall. Convert the height of the building to:

- | | |
|---------------------|----------------------|
| a Kilometres | b Centimetres |
|---------------------|----------------------|

10 Neville is 141 cm tall and Rosey is 1.78 m tall.

- a** Convert Rosey's height from metres to centimetres.
- b** Who is shorter, Rosey or Neville?

11 Plant 1 grows 4.8 cm in a week, while Plant 2 grows 50 mm in the same week. Which plant grows more in the week?

Area

12 Convert the following:

- | | |
|---|--|
| a 3 m ² to square centimetres (cm ²) | b 7 m ² to square centimetres (cm ²) |
| c 0.56 km ² to square metres (m ²) | d 54 800 m ² to hectares (ha) |
| e 810 mm ² to square centimetres (cm ²) | f 10.4 ha to square metres (m ²) |
| g 34 000 cm ² to square metres (m ²) | h 5.73 cm ² to square millimetres (mm ²) |

13 Convert the following:

- | | |
|---|---|
| a 790 000 000 cm ² to square kilometres | b 6 ha to square metres |
| c 20 000 m ² to hectares | d 8 750 m ² to hectares |

14 Find the area of a square of side 290 cm, in square metres.

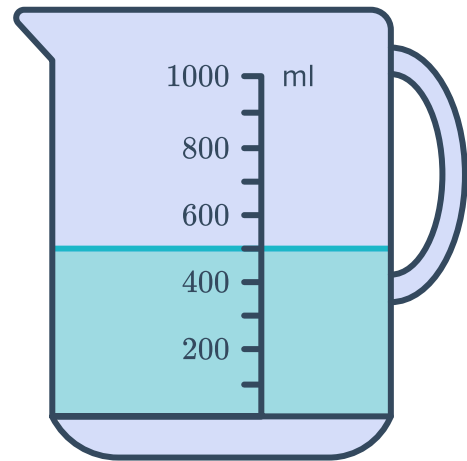
15 A park has an area measuring 19 acres. Given that 1 acre is approximately 4000 m², find the



- 16** A property covers an area of 4.5 ha. Given that 1 ha is $10\,000\text{ m}^2$, calculate the following:
- a** The area of the property in square metres.
 - b** The area of the property in acres, given that 1 acre is approximately 4000 m^2 .
- 17** A large cattle station covers an area of 320 acres. Given that 1 acre is approximately 0.4 ha, and 1 ha is $10\,000\text{ m}^2$, calculate the following:
- a** The area of the cattle station in hectares.
 - b** The area of the cattle station in square kilometres.

Capacity

- 18** What metric unit represents 'one thousandth of a litre'?
- 19** How much water is in the jug shown? Give your answer in litres.



- 20** Convert the following:

- | | |
|-------------------------------------|---|
| a 2000 mL to litres (L) | b 4920 mL to litres (L) |
| c 240 mL to litres (L) | d 5 kL to litres (L) |
| e 1.67 kL to litres (L) | f 0.4 L to millilitres (mL) |
| g 1.54 L to millilitres (mL) | h $3\frac{4}{5}$ L to millilitres (mL) |
| i 84.6 L to kilolitres (kL) | j 1425 L to kilolitres (kL) |

- 21** James drank $\frac{9}{10}$ litres of water. How much did he drink in millilitres?



- 23** 279 children went on a picnic. Each child drank 400 millilitres of orange juice. Calculate the number of litres consumed altogether.